

Project Title: FeedInsight – AI Product Intelligence & Backlog Automation

Team Name: FeedInsight Team

Target Industry: Tech

Team Members:

1. Abdelrahman Reda Mohammed
2. Abdelrahman Nasr Aql
3. Alaa Essam Ali
4. Mostafa Shaaban Mohamed
5. Ahmed Essam Hamdy
6. Ali Azouz Ali
7. Yousef Hanna Anees

The Core Problem (The "Why")

Product Owners and engineering teams often receive hundreds of customer feedback reports for each product release, making manual triage slow, repetitive, and difficult to scale.

For example:

- Customers often combine multiple requests in a single feedback. For instance, a customer may report that the checkout page freezes when paying with Visa while also requesting Apple Pay support. Although it is a single feedback, it represents two different engineering tasks: a bug report and a feature request.
- Customers describe problems using natural, non-technical language, while developers need clear, structured technical requirements. This creates a gap between customer feedback and actionable engineering work that requires manual interpretation.
- Similar issues are often reported using different wording, making it difficult to detect duplicate tickets and prioritize the real problems.
- As customer feedback accumulates, it becomes increasingly difficult to identify recurring issues, understand overall customer sentiment, and discover the most requested features without manually reviewing hundreds of reports.
- Once feedback moves into the Development stage, it needs to remain connected to the engineering workflow. Developers need to be able to review the relevant customer feedback and

respond with comments or updates, while customers should remain connected to the feedback they originally submitted.

Core Insight

The real challenge is not collecting customer feedback, but transforming large volumes of unstructured feedback into actionable product insights, developer-ready backlog items, and an organized workflow that keeps customers and engineering teams connected throughout the development process.

2. The Solution & Value Proposition (The "What")

FeedInsight is an AI-powered Product Intelligence platform that helps Product Owners and engineering teams understand customer feedback, gain actionable insights, and automate backlog triage.

The platform intelligently analyzes customer feedback, separates multiple issues within a single submission, detects customer sentiment, identifies recurring trends and feature requests, searches the existing backlog for related tickets, and prepares developer-ready tickets for human review before they are added to the production backlog.

FeedInsight also provides interactive dashboards for visualizing product insights and an AI Assistant that enables teams to explore feedback and product data using natural language.

Once feedback moves into the **Development stage**, FeedInsight keeps the original customer feedback connected to the engineering workflow, allowing developers to review relevant feedback and add comments or updates while keeping the customer connected to their submitted feedback.

Key Value:

Helping product teams transform large volumes of customer feedback into actionable insights, data-driven product decisions, and developer-ready backlog items, while reducing manual triage effort, minimizing duplicate tickets, and maintaining a connected feedback-to-development workflow.

3. Platform Overview

Customer Portal:

- **Submit Customer Feedback:** Customers can submit feedback with optional screenshots and additional details to provide context about their issue or request.
- **Feedback Tracking:** Customers can view the feedback they have submitted and follow its progress as it moves through the product workflow, including when it reaches the Development stage.
- **Feedback Comments:** When feedback reaches the Development stage, customers can view comments or updates added by the engineering team while keeping the discussion connected to their original feedback.

Admin Portal:

- **AI Analytics Dashboard:** Provides an interactive overview of customer feedback, including customer sentiment, recurring issues, most requested features, feedback trends, and AI-generated ticket statistics to support data-driven product decisions.
- **Feedback Management:** Allows Product Owners to browse submitted feedback, review AI-generated analysis (intent, sentiment, and category), search and filter feedback, and inspect related or duplicate reports.
- **AI Product Assistant:** Enables Product Owners to ask natural-language questions about customer feedback, recurring issues, feature requests, and existing backlog items, providing instant AI-generated insights powered by the platform's AI knowledge base.
- **Backlog Review:** Presents AI-generated draft tickets, duplicate matches, suggested priorities, and developer-ready User Stories with Acceptance Criteria. Product Owners can review, edit, approve, or reject AI suggestions before they are added to the Jira backlog.

Developer Portal:

- **Development Feedback:** Allows Developers to view customer feedback that has reached the **Development stage**, providing the original customer context alongside the related engineering work.

- **Developer Comments & Updates:** Developers can add comments or updates directly to the relevant customer feedback, keeping technical progress and customer context connected within the same workflow.
-

4.The AI Trinity Strategy (The "How")

• LLM (Intelligence)

We will use GPT-4o mini because it provides fast responses, excellent cost efficiency, and is capable of generating reliable structured outputs required by our workflow.

The LLM is responsible for:

- Analyze unstructured customer feedback and bug reports.
- Detect one or more intents within a single feedback.
- Extract core intents and translate customer language into technical domain keywords.
- Analyze customer sentiment (Positive, Neutral, or Negative).
- Evaluate the retrieved knowledge to accurately detect duplicated or overlapping issues.
- Generate concise summaries of recurring customer feedback for dashboards and AI Assistant responses.
- Generate developer-ready User Stories and explicit Acceptance Criteria (Given-When-Then) for new tickets.

• RAG (Knowledge)

To provide context-aware insights and accurate engineering decisions, our RAG pipeline continuously synchronizes the platform's AI knowledge base.

Knowledge Sources:

- Customer feedback submitted through the platform.
- Existing Jira product backlog.
- **Continuous Ingestion:**Jira Webhooks automatically trigger an event whenever a developer creates or updates a ticket in the backlog. Customer feedback submitted through the platform is also continuously synchronized.

- **Embedding:**Raw customer feedback and Jira ticket data are converted into vector embeddings using **OpenAI text-embedding-3-small** and stored in a **Qdrant Vector Database**. Each Jira ticket and each customer feedback entry are treated as a single semantic chunk for semantic retrieval.
 - **Semantic Retrieval:**When new customer feedback arrives or the AI Assistant receives a user query, the request is embedded and queried against **Qdrant** to retrieve the Top-K most relevant customer feedback and backlog items before passing them to the LLM.
 - **Context Generation:**
The retrieved knowledge is used by the LLM to generate:
 - Product insights.
 - Dashboard summaries.
 - AI Assistant responses.
 - Developer-ready backlog tickets.
-

• Agents (Action)

To orchestrate these agents, the platform utilizes **Semantic Kernel** as its primary agent framework.

The system processes customer feedback step by step:

1. Router Agent

Responsibilities:

- Reads the raw customer feedback.
- Detects one or more intents within the feedback.
- Splits compound feedback into separate tasks when multiple issues are detected.
- Classifies each task (e.g., Bug Report or Feature Request).
- Extracts technical keywords for semantic retrieval.
- Identifies customer sentiment (Positive, Neutral, or Negative).

Tools Used:

- No external tools.

2. Triage Agent

Responsibilities:

- Compares the processed feedback against existing tickets retrieved from the vector database.
- Searches for semantically similar backlog items.
- Decides whether the issue is a duplicate, should be linked to an existing ticket, or represents a completely new problem.
- Generates a clear, developer-ready User Story with Acceptance Criteria for new issues.

Tools Used:

- **Gmail Tool:** Dispatches automated email notifications to Product Owners when a critical bug or high-priority issue is detected.
- **Qdrant Tool:** Runs semantic queries against the vector database to retrieve existing tickets and evaluate duplicates.
- **Jira Tool:** Automatically formats and creates draft User Stories and Acceptance Criteria for Product Owner review before adding them to the product backlog.

Agent Boundaries:

The AI agents **can**:

- Analyze customer feedback.
- Search the backlog semantically.
- Detect duplicate issues.
- Generate draft developer-ready tickets.
- Generate product insights for dashboards and AI Assistant responses.

The AI agents **cannot**:

- Modify the production backlog directly.
- Delete or close Jira tickets.
- Override Product Owner decisions.

5. Market & Competition

Target Market:

- Software companies

- SaaS companies
- Enterprise software teams
- Product-driven startups

Primary Users:

- Product Owners
- Product Managers
- Engineering Managers

The Advantage:

Traditional tools such as **Jira, Zendesk, and Intercom** help teams collect customer feedback and manage support requests. However, Product Owners still spend significant time manually interpreting customer feedback, searching for duplicate issues, and converting customer language into developer-ready backlog items.

FeedInsight adds an AI-powered intelligence layer on top of the existing engineering workflow.

Unlike traditional tools, FeedInsight:

- Understands customer feedback semantically rather than relying on keywords.
- Splits compound feedback into multiple engineering tasks.
- Detects duplicate or related issues using semantic search.
- Generates developer-ready User Stories and Acceptance Criteria.
- Integrates with the existing product backlog to support faster and more consistent triage.
- Provides AI-powered dashboards to visualize customer sentiment, recurring issues, and feature requests.
- Enables Product Owners to interact with an AI Assistant using natural language.
- Helps teams discover actionable product insights from customer feedback.

This allows Product Owners to spend less time on manual triage while making faster, data-driven product decisions.

6. Execution Plan (High Level)

Core Stack

- **Frontend:** React
- **Backend:** ASP.NET Core Web API
- **Database:** SQL Server
- **AI Model:** GPT-4o mini
- **AI Framework:** Microsoft Semantic Kernel
- **Vector Database:** Qdrant
- **Embedding Model:** OpenAI text-embedding-3-small

Deployment

- **CI/CD:** GitHub Actions
- **Hosting Provider:** MonsterASP
-

• Data Privacy & Security

To protect customer and business data, the platform will:

- Use JWT-based authentication and role-based authorization.
- Encrypt all communication using HTTPS.
- Store only the necessary ticket data for semantic search.
- Restrict access to product backlog data based on user permissions.

7. Risk Factors

- **Duplicate Matching Accuracy** :Ensuring the AI correctly distinguishes between duplicate issues and genuinely new customer requests.
- **Requirement Generation Quality** :Ensuring generated User Stories and Acceptance Criteria are technically accurate, complete, and useful for developers.
- **Keeping the Knowledge Base Up-to-Date** :Maintaining synchronization between the Jira backlog and the vector database to ensure accurate semantic retrieval.
- **Insight Quality** :Ensuring AI-generated summaries and dashboard insights accurately reflect customer feedback without introducing misleading conclusions.